Olist Marketing Funnel & Lead Conversion Analysis

From marketing-qualified lead to closed deal: where leads convert and where they are lost

Tools: Python ● Pandas ● SQL ● Excel (PivotTables, formulas, charts)

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1. Company Background

Olist is a Brazilian e-commerce company that helps small and medium businesses sell their products through online marketplaces. To grow its network of sellers, Olist runs a **sales funnel**: marketing attracts interested merchants, and a sales team turns them into active sellers.

This project analyses the public **Olist Marketing Funnel dataset** (Kaggle): 8,000 marketing-qualified leads (MQLs) who first made contact between June 2017 and May 2018, and the 842 of them who became closed deals (new sellers).

2. Business Problem

Only about **1 in 10** marketing-qualified leads becomes a closed deal. The sales and marketing teams need to know **where leads are being lost, which lead sources convert, and how long deals take to close**, so that marketing budget and sales effort go to the right places.

The analysis answers three questions:

- **How many** leads convert, and how does this change over time and by lead source?
- **How long** does it take to close a deal?
- **What do** the closed deals look like (lead type, segment, behaviour profile, sales team)?

3. Business Objectives

#	Objective	Measured by	Result
1	Measure MQL to closed-deal conversion overall, by origin and by month cohort	Conversion rate %	10.5% overall; 3.0% (Jun-Dec 2017) vs 13.0% (Jan-May 2018)
2	Find which lead origins convert well or badly	Conversion rate and share of MQLs vs closed deals	Organic and paid search about 12%; social 5.6%; email 3.0%
3	Measure sales-cycle length	Days from first contact to won date	Median 14 days, average 48.5, 17.8% over 90 days
4	Profile the closed deals	Share of closed deals by lead type, segment, behaviour, SDR, SR	online_medium 39.4%; top 5 SDRs 49.9%
5	Build an Excel dashboard and give recommendations	8 KPI cards, 6 charts, 6 PivotTables, 18 self-checks, 7 recommendations	Delivered

4. Stakeholders

Stakeholder	What they need to decide	Dashboard view used
Head of Sales / Sales Manager	Where to focus follow-up; workload across reps	Funnel, days to close, SDR/SR tables
Marketing Manager	Which channels deserve budget	Lead origin chart and origin table
SDR and SR team leads	Follow-up timing and lead allocation	Time to close, closed deals by lead type
Sales / BI Operations	Data quality and tracking	Origin tracking (unknown / blank origin)
Senior Management	Overall funnel health	KPI cards and monthly cohort chart

5. Business KPIs

KPI	Formula	Value
Total MQLs	Count of marketing-qualified leads	8,000
Closed Deals	MQLs with a matching closed-deal record	842
Conversion Rate (all months)	Closed Deals ÷ Total MQLs. Blends two periods that are not comparable	10.5% (10.525%)
Conversion Rate, Jan-May 2018	Same formula on the comparable window (781 of 5,998). Use as the benchmark	13.0%
Non-converted leads	Total MQLs − Closed Deals	7,158
Drop-off	Non-converted ÷ Total MQLs	89.5%
Average days to close	Mean of (won date − first contact date)	48.5
Median days to close	Middle value of the same measure	14.0
Closed within 30 days	Deals closed in 30 days or less ÷ deals with a valid date gap (554 of 841)	65.9%
Unique SDRs / Sales Reps	Distinct IDs among closed deals	32 / 22

Use the median (14 days), not the average (48.5), as the typical sales cycle. A small number of very slow deals pull the average up.

6. Domain Knowledge

Term	Meaning
MQL (Marketing Qualified Lead)	A lead that marketing considers ready for the sales team, here a merchant who showed interest through a landing page.
SDR (Sales Development Representative)	Qualifies the lead and holds the first conversations.
SR (Sales Representative)	Closes the deal. The converted merchant becomes a seller.
Funnel	The stages a lead passes through. This dataset covers only the first and last: MQL → closed deal. Intermediate stages (contacted, consultancy) are not in the data.
Conversion rate	Share of leads that reach the next stage. Always read it together with volume.
Lead origin	The marketing channel the lead came from (organic search, paid search, social, email ...).
Cohort	A group of leads that first made contact in the same month, followed over time.
Sales cycle	Time from first contact to winning the deal.
Attribution	Linking a conversion to its source. Missing or "unknown" origin weakens it.

7. Dataset Overview

Item	Detail
Source	Olist Marketing Funnel dataset (Kaggle). Two CSV files.

Item	Detail
MQL table	8,000 rows × 4 columns: mql_id, first_contact_date, landing_page_id, origin. First contact 14 Jun 2017 to 31 May 2018.
Closed deals table	842 rows × 14 columns. Won dates 5 Dec 2017 to 14 Nov 2018.
Join	LEFT JOIN of closed deals onto MQLs on mql_id (one to one). A lead is "Converted" if a closed-deal record exists. Every closed deal exists in the MQL table.
Duplicates	0
Missing values	MQL origin: 60 (0.75%). Closed deals: behaviour profile 177 (21%); company, GTIN, stock and catalog-size questionnaire fields about 92% missing.
Other data issues	1 deal with a won date before its first-contact date (excluded from days-to-close); 797 of 842 declared monthly revenue values are 0 and one is 50,000,000, so revenue is not used.

8. Data Dictionary

Column	Table	Description
mql_id	Both	Unique lead ID; join key
first_contact_date	MQL	Date the lead first made contact (start of the sales clock)
landing_page_id	MQL	Landing page the lead signed up on
origin	MQL	Marketing channel (organic_search, paid_search, social, ...)
seller_id	Closed	Seller account created after the deal was won
sdr_id, sr_id	Closed	Sales Development Rep and Sales Rep on the deal
won_date	Closed	Date-time the deal was won (end of the sales clock)
business_segment	Closed	Product category the seller will sell
lead_type	Closed	Kind of seller (online_medium, industry, offline ...)
lead_behaviour_profile	Closed	Behaviour persona (cat, eagle, wolf, shark; some have two)
has_company, has_gtin, average_stock, declared_product_catalog_size	Closed	Questionnaire answers, filled for only about 8% of deals. A blank means not answered, not "No".
business_type	Closed	reseller / manufacturer / other
declared_monthly_revenue	Closed	Declared monthly revenue (BRL). Mostly 0, not used.
Created: conversion_status, converted_flag, days_to_close, first_contact_month, won_day	Master	Analytical columns built in Python

9. Business Questions

Business question	Answer from the data
What share of leads becomes sellers, and is it improving?	10.5% overall. Jan-May 2018 cohorts convert at 13.0%; earlier cohorts are not comparable (first deal won Dec 2017).
Which channels bring leads that actually convert?	Organic and paid search: about 12%. Social 5.6% and email 3.0% are far below the 10.5% average.
Is any channel under-reported?	14.5% of MQLs have unknown or blank origin, so the true channel mix is unclear.
How long does a deal take?	Median 14 days; 65.9% close within 30 days; 17.8% take over 90.

Business question	Answer from the data
What do closed deals look like?	online_medium is 39.4%; top 5 segments are 49.3%; "cat" profile is 48.3%.
Which profiles close slowly?	Wolf: median 36 days vs 13 for cat.
Is the sales workload balanced?	No. The top 5 SDRs handled 49.9% and the top 5 SRs 49.4% of closed deals.

10. Data Cleaning & Preparation

Cleaning was done in a Google Colab notebook with Python and Pandas. **No rows were deleted**; every decision is documented.

Issue	Decision	Reason
Text values with different case or spaces	Trimmed and lower-cased	Avoid duplicate categories
first_contact_date is a date, won_date is a date-time	Compared on the date only (won_day)	Like-for-like day count
60 MQLs with blank origin	Labelled "not_provided"; kept separate from the existing "unknown"	Blank and unknown may mean different things
177 closed deals without behaviour profile	Labelled "Not Provided"; never guessed	Do not invent a persona
About 92% of questionnaire fields blank	Kept as NULL; not used in KPIs	Blank = not answered, not "No"
Revenue mostly 0 with one extreme value	Not used	Zero most likely means not declared
1 deal won before first contact	days_to_close set to blank; row kept	Data error; keep the lead in conversion counts
16 deals with two behaviour profiles (for example "cat, wolf")	Flagged, kept as given	Do not split without business rules

A master table was built as MQL LEFT JOIN closed deals with conversion_status, days_to_close and first_contact_month, and exported to Excel for PivotTables.

11. Python & SQL Analysis

Python (Pandas) was used for cleaning, the master table, KPIs and the Excel export. **SQL** was used to answer the same business questions from a database: four scripts in the sql folder (create tables, load data, quality checks with an analysis view, and 12 analysis queries). Every query result is saved in data/query_results and matches the Python and Excel numbers.

SQL script	What it does
01_create_database_and_tables.sql	Creates the database and two tables (mql, closed_deals) with keys and indexes
02_load_data.sql	Loads the two cleaned CSV files; checks 8,000 and 842 rows
03_data_quality_and_cleaning.sql	8 data-quality checks, then builds the lead_master view (one row per lead) and a control query
04_analysis_queries.sql	12 business queries: funnel, comparable window, monthly cohort, origin, channel groups, landing pages, time to close, lead type, segments, behaviour, SDR workload

SQL technique	Business question	Result
Aggregates on a LEFT JOIN view	How many leads sign up?	8,000 leads; 842 closed; 10.5%
GROUP BY cohort period	Is 10.5% a fair headline?	Jun-Dec 2017: 3.05%; Jan-May 2018: 13.02%
CASE WHEN for 30-day conversion	Do months compare fairly?	Jan-Apr 2018: 8.4% to 11.0% within 30 days; May: 7.1%
GROUP BY + conditional SUM	Which channel converts?	Search about 12%; social 5.56%; email 3.04%
CASE channel groups	How do channel groups compare?	Social + email: 23.0% of leads, 10.7% of deals

SQL technique	Business question	Result
CTE + ROW_NUMBER (median)	Typical time to close?	Median 14 days; average 48.5
GROUP BY + LIMIT	Biggest landing pages	Two big weak pages are 93-94% social traffic
Subquery share	Closed deals by SDR	Top 5 SDRs = 49.9% of deals

Conversion by lead origin

Origin	MQLs	Closed	Conversion	% of MQLs	% of closed	Jan-May 2018 only
organic_search	2,296	271	11.80%	28.7%	32.2%	14.7%
paid_search	1,586	195	12.30%	19.8%	23.2%	14.8%
social	1,350	75	5.56%	16.9%	8.9%	6.3%
unknown	1,099	179	16.29%	13.7%	21.3%	20.7%
direct_traffic	499	56	11.22%	6.2%	6.7%	14.2%
email	493	15	3.04%	6.2%	1.8%	4.0%
referral	284	24	8.45%	3.5%	2.9%	11.4%
other	150	4	2.67%	1.9%	0.5%	3.5%
display	118	6	5.08%	1.5%	0.7%	6.7%
other_publicities*	65	3	4.62%	0.8%	0.4%	7.3%
not_provided*	60	14	23.33%	0.8%	1.7%	27.1%

* fewer than 100 leads: read with caution.

Monthly cohort (grouped by first-contact month)

Month	MQLs	Closed	Conversion (ever)	Closed within 30 days	Conversion (30 days)
2017-06	4	0	0.00%	0	0.00%
2017-07	239	2	0.84%	0	0.00%
2017-08	386	9	2.33%	0	0.00%
2017-09	312	7	2.24%	0	0.00%
2017-10	416	14	3.37%	0	0.00%
2017-11	445	18	4.04%	2	0.45%
2017-12	200	11	5.50%	1	0.50%
2018-01	1,141	152	13.32%	96	8.41%
2018-02	1,028	149	14.49%	101	9.82%
2018-03	1,174	167	14.22%	114	9.71%
2018-04	1,352	183	13.54%	148	10.95%
2018-05	1,303	130	9.98%	92	7.06%

Read the cohort table carefully. The first deal in the data was won in Dec 2017 and MQL volume jumps from 200 (Dec) to 1,141 (Jan). Jun-Dec 2017 cohorts convert at 3.0% (61 of 2,002) and Jan-May 2018 at 13.0% (781 of 5,998). The early months are **not comparable** with the later ones, so the rise is not proof of improvement. Jan 2018 onward is the better benchmark, and the 30-day column gives every month the same time to close.

Time to close

Days to close	Closed deals	Share
0-7 days	283	33.7%
8-30 days	271	32.2%
31-90 days	137	16.3%
Over 90 days	150	17.8%

12. Excel Dashboard

The dashboard workbook (Olist_Funnel_Dashboard.xlsx) is built in Excel. Every number on it is a live formula on the data sheet, the six PivotTables are native Excel PivotTables, and a Checks sheet proves the numbers tie out.

Sheet	What it contains
Dashboard	8 KPI cards, 6 charts, an origin table with colour scale and arrows, a top-5 landing-page table and a live "ALL CHECKS PASS" footer
Insights	11 findings with live-formula evidence, 7 recommendations and the limitations
Analysis	All tables behind the charts (live formulas, no pasted values)
PivotTables	6 native PivotTables: origin, month, lead type, segment, SDR and cohort period
Checks	18 self-tests: row counts, totals tie out, pivot totals, KPIs vs the notebook
Master_Data, MQL_Data, Closed_Deals, Data_Notes	All the data, the data dictionary and the cleaning decisions

- **KPI cards:** Total MQLs, Closed Deals, Conversion Rate (all months), **Conversion Rate Jan-May 2018 (comparable benchmark)**, Not Converted, Average days, Median days, Closed within 30 days.
- **Charts:** funnel; monthly cohort (MQLs, conversion % ever and within 30 days, with a caveat that months before 2018 are not comparable); lead origin (volume and conversion %, * = under 100 leads); closed deals by lead type; days to close; top 10 segments.
- **Built-in cautions:** origins under 100 leads are starred, closed-deal views are labelled as counts and not conversion rates, and the unknown-origin group is flagged as a tracking gap.

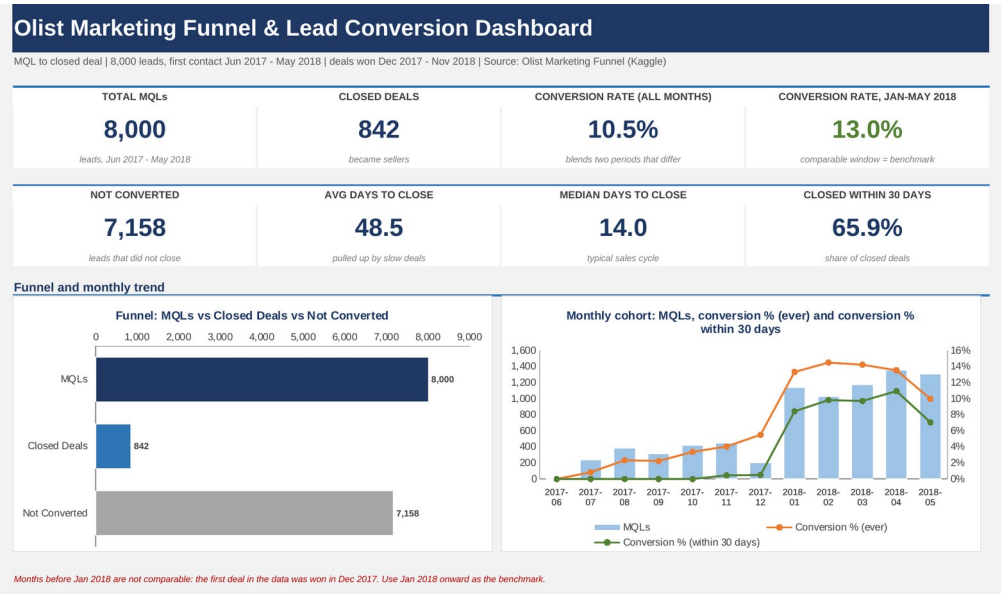


Figure 1: Excel dashboard (top): KPI cards, funnel and monthly cohort

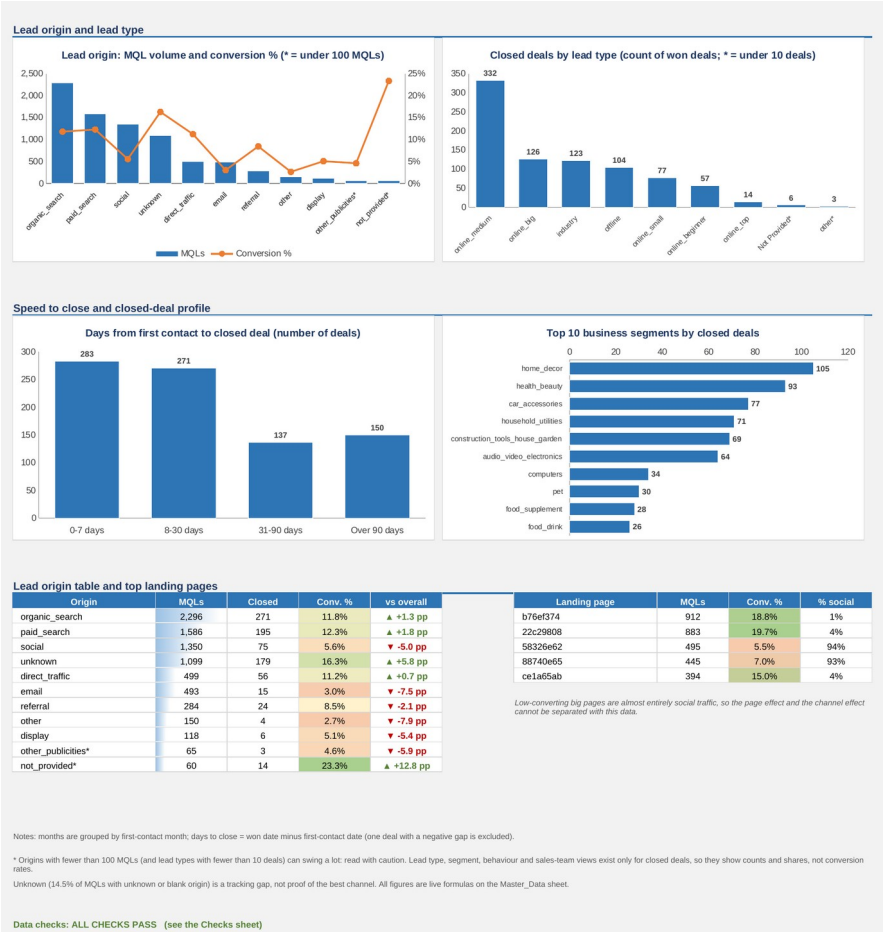


Figure 2: Excel dashboard (lower): lead origin, lead type, speed to close, segments and tables

13. Key Business Insights

#	Insight	Evidence
1	Only about 1 in 10 leads becomes a closed deal	842 of 8,000 = 10.5%; 7,158 not converted. Intermediate stages are not in the data, so the stage where leads leak cannot be identified.

#	Insight	Evidence
2	Early months are not comparable with later months	Jun-Dec 2017: 61 of 2,002 = 3.0%. Jan-May 2018: 781 of 5,998 = 13.0%. First deal won Dec 2017; MQL volume 200 → 1,141. The 10.5% headline blends both periods.
3	Organic and paid search are the core channels	48.5% of MQLs and 55.3% of closed deals; conversion 11.8% and 12.3%.
4	Social and email have volume but low conversion	Social 1,350 MQLs at 5.6%; email 493 at 3.0%. Together 23.0% of MQLs but 10.7% of closed deals. Restricted to Jan-May 2018 the gap remains (social 6.3%, email 4.0% vs about 14.7% for organic and paid search).
5	Origin tracking has a gap	1,159 MQLs (14.5%) have unknown or blank origin; 193 closed (16.7%). This is a tracking problem, not evidence of the best channel.
6	The sales cycle has a long tail	Median 14 days vs average 48.5. 65.9% close within 30 days; 17.8% take over 90 days.
7	A few lead types make most closed deals	online_medium 332 deals (39.4%); top 3 types 69.0%. This is a share of won deals, not a conversion rate.
8	Closed deals concentrate in a few segments	Top 5 segments = 49.3% (home_decor 105, health_beauty 93).
9	Wolf-profile deals take longer	Median 36 days vs 13 (cat) and 10 (eagle); 21.0% of deals have no profile.
10	Closed deals are concentrated among a few reps	32 SDRs and 22 SRs; top 5 SDRs = 49.9%, top 5 SRs = 49.4%. Leads worked per person are not in the data, so this is workload, not performance.

Every figure above is reproduced by a live formula in the Insights sheet of the Excel workbook and by the SQL queries.

14. Business Recommendations

These are suggestions based on the observed patterns. They are **not guaranteed** to raise conversion and should be tested before changing budgets.

#	Recommendation	Based on	Owner	Success measure
1	Review lead quality and follow-up for social and email before adding budget	Insight 4	Marketing Manager	Conversion % by channel each month
2	Keep organic and paid search as core channels; compare channels on volume and conversion together	Insight 3	Marketing Manager	Share of closed deals; cost per closed deal (needs spend data)
3	Improve origin tracking so unknown and blank origins can be attributed	Insight 5	Sales / BI Operations	% of MQLs with unknown or blank origin (now 14.5%)
4	Use Jan 2018 onward as the benchmark; confirm whether deal recording changed around Dec 2017; compare cohorts on a fixed window such as 30-day conversion	Insight 2	BI team with data owners	Cohort conversion on a like-for-like window
5	Set a follow-up checkpoint around 30 days; review 90+ day deals and slower profiles such as wolf separately	Insights 6, 9	Sales Manager	% closed within 30 days (now 65.9%); share over 90 days (now 17.8%)
6	Capture lead type and segment at the MQL stage	Insights 7, 8	Sales Operations	Conversion rate by lead type and segment becomes measurable
7	Check how deals are spread across SDRs and SRs; record leads worked per rep	Insight 10	Sales Manager	Conversion per rep; allocation balance

15. Business Impact (illustrative)

The data has no revenue per deal and no marketing spend (declared revenue is 0 for 95% of deals), so a financial return cannot be calculated. The scenario below uses **deal counts only** and is **not a forecast**.

In the comparable Jan-May 2018 window, social and email brought 1,398 MQLs and 80 closed deals (5.7%), while organic and paid search converted at 14.7%. If social and email conversion rose to:

Assumed conversion for social + email	Extra closed deals (5 months)	Increase on the 781 deals in the window
8%	about 32	about +4%
10%	about 60	about +8%
12%	about 88	about +11%

Caution: these channels may not be able to reach these rates, and improving lead quality has a cost. Use the scenario to size the opportunity, not to promise a result.

Other benefits: one agreed set of funnel KPIs, earlier detection of conversion drops, and a clear list of data to capture (leads worked per rep, stage dates, marketing spend).

16. Limitations

- Only two funnel stages (MQL and closed deal) exist. Contacted and consultancy stages are not in the data, so the stage where leads leak cannot be identified.
- Lead type, segment, behaviour profile, SDR and SR exist only for closed deals, so those views show shares of won deals, not conversion rates, and reps cannot be ranked.
- The first deal was won in Dec 2017 and MQL volume changes sharply in Jan 2018, so months before 2018 are not comparable. Cohorts that first made contact later have had less time to close deals.
- About 92% of questionnaire fields are blank and revenue is mostly 0, so these fields are not used.
- The unknown and blank origin group (14.5% of MQLs) limits channel attribution.
- No marketing cost or revenue per deal, so no cost-per-deal or ROI. Origin differences are observational and do not prove cause.
- landing_page_id is available for every lead but was not analysed. It is closely tied to origin (the two largest low-converting pages are about 94% social traffic), so any landing-page analysis must be read together with origin.

17. Business Analyst Deliverables

17.1 Business requirement summary (BRD)

Item	Detail
Business need	A trusted, repeatable view of lead conversion, lead-source performance and sales-cycle length.
Scope (in)	Conversion analysis from MQL to closed deal in Python and SQL, an Excel dashboard, insights and recommendations.
Scope (out)	Prediction models, marketing cost and revenue analysis, rep performance ranking.
Success criteria	KPIs agree between Python, SQL and Excel; stakeholders can answer the 7 business questions from the dashboard.

Item	Detail
Assumptions	A lead is converted if a closed-deal record exists; won date is the end of the sales clock; origin values are as recorded.

17.2 Functional requirements (FRD summary)

ID	Requirement
FR-1	Dashboard shall show 8 KPI cards: total MQLs, closed deals, conversion rate (all months), conversion rate Jan-May 2018, non-converted, average days, median days and % closed within 30 days.
FR-2	Dashboard shall show the funnel and the monthly cohort (MQLs, closed deals, conversion %).
FR-3	Dashboard shall show MQL volume and conversion % by lead origin, with a comparison to the overall rate.
FR-4	Dashboard shall show closed deals and average days to close by lead type.
FR-5	Views based on closed deals only shall state that they are counts or shares, not conversion rates.

17.3 User stories and acceptance criteria

User story	Acceptance criteria
As a Marketing Manager, I want conversion by lead origin so that I can decide which channels to fund.	Origin chart and table show all origins; social shows 5.56% and paid search 12.30%.
As a Sales Manager, I want the typical time to close so that I can set a follow-up checkpoint.	Median 14.0 and average 48.5 days are shown; time bands add to 842 deals less one excluded.
As Senior Management, I want one page of funnel KPIs so that I can check health quickly.	KPI cards show 8,000, 842, 10.5%, 13.0%, 7,158, 48.5, 14.0 and 65.9%.
As a BI analyst, I want caveats shown so that nobody misreads the cohort trend.	Notes and the Insights sheet state that early months are not comparable.

17.4 As-Is vs To-Be

	As-Is	To-Be
Funnel view	Lead and deal counts are not tied together.	One master table links every MQL to its outcome.
Channel decisions	Judged by lead volume.	Judged by volume and conversion together.
Sales cycle	Average only, distorted by slow deals.	Median and time bands with a 30-day checkpoint.
Data	Unknown origin and missing stage data.	Origin tracking and stage dates captured.

To-Be process flow: MQL and closed-deal data → cleaning and join → KPIs and cohorts → Excel dashboard → monthly review → channel and follow-up actions → re-measure conversion.

17.5 Stakeholder matrix

Stakeholder	Influence	Interest	How to engage
Head of Sales	High	High	Monthly funnel review
Marketing Manager	High	High	Channel results and tracking fixes
SDR / SR team leads	Medium	High	Follow-up timing and allocation
BI / Data team	Medium	High	Owns data, KPIs and dashboard
Senior Management	High	Medium	Summary KPIs

17.6 Risk register

Risk	Impact	Mitigation
Early cohorts read as real improvement	Wrong conclusion about performance	Annotate cohort chart; use Jan 2018 onward as benchmark
Small origins show extreme rates (for example not_provided, n = 60)	Over-reading noise	Flag origins under 100 MQLs
Closed-deal views read as conversion rates	Wrong segment decisions	Label as counts and shares
Unknown origin treated as best channel	Misdirected budget	Fix tracking before using it
SDR/SR shares read as performance	Unfair people decisions	Record leads worked per rep first
Pivot tables not refreshed	Pivot view shows stale values	Pivots refresh on open; dashboard uses live formulas; Checks sheet reconciles pivot totals

18. Conclusion, Tools & Deliverables

About 1 in 10 Olist leads becomes a closed deal. Organic and paid search are the strongest sources at about 12% conversion, while social (5.6%) and email (3.0%) bring volume but convert poorly. Half of deals close within two weeks, but a long tail of slow deals pulls the average up to 48.5 days. Two cautions matter for decisions: months before 2018 are not comparable with later ones, and lead type, segment and sales-team views describe won deals only, not conversion. The recommendations are based on observed patterns and should be tested before changing budgets.

Tools & technologies

- **Python (Google Colab):** Pandas, NumPy for cleaning, joins, KPIs, cohorts and exports
- **SQL (MySQL 8 syntax):** LEFT JOIN view, CASE, GROUP BY, CTEs, window functions (ROW_NUMBER) and subqueries
- **Excel:** native PivotTables, live formulas, charts, conditional formatting and a self-checking Checks sheet

Project deliverables

- python/Olist_Funnel_Analysis_Colab.ipynb (Python and SQL notebook)
- sql/01 to 04 (MySQL scripts: tables, load, quality checks, 12 analysis queries)
- excel/Olist_Funnel_Dashboard.xlsx (dashboard, 6 PivotTables, Checks) and dashboard_screenshot.png
- data/raw, data/cleaned, data/query_results and data_dictionary.csv
- This project report and the GitHub README